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The Influence of Movement Tempo on Acute Neuromuscular, Hormonal, and Mechanical Responses to Resistance Exercise—A Mini Review

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Abstract

Wilk, M, Tufano, JJ, and Zajac, A. The influence of movement tempo on acute neuromuscular, hormonal, and mechanical responses to resistance exercise—a mini review. *J Strength Cond Res* 34(8): 2369–2383, 2020—Resistance training studies mainly analyze variables such as the type and order of exercise, intensity, number of sets, number of repetitions, and duration and frequency of rest periods. However, one variable that is often overlooked in resistance training research, as well as in practice, is premeditated movement tempo, which can influence a myriad of mechanical and physiological factors associated with training and adaptation. Specifically, this article provides an overview of the available scientific literature and describes how slower tempos negatively affect the 1-repetition maximum, the possible load to be used, and the number of repetitions performed with a given load, while also increasing the total time under tension, which can mediate acute cardiovascular and hormonal responses. As a result, coaches should consider testing maximal strength and the maximal number of repetitions that can be performed with each movement tempo that is to be used during training. Otherwise, programming resistance training using various movement tempos is more of a trial-and-error approach, rather than being evidence or practice based. Furthermore, practical applications are provided to show how movement tempo can be adjusted for a variety of case study–type scenarios.

Key Words: repetitions, time under tension, 1RM test, testosterone, cortisol, growth hormone

Introduction

Resistance training is a common form of exercise for athletes and other types of physically active people. When designing a resistance training program, there are several acute training variables that can be manipulated to guide the direction and scope of the desired adaptive changes (7,62). Of these acute training variables, the most commonly investigated include exercise intensity, which often expressed as a percentage of 1-repetition maximum (% 1RM) and volume, which is determined by the number of sets and repetitions completed during training (10,53). However, the duration and frequency of rest periods also play a major role in the process of acute resistance training programming (10). Therefore, scientists must investigate a large number of possible intensity-volume-rest period combinations, and the scientific literature widely covers the effects of various combinations of these variables. However, the movement tempo of individual repetitions is another variable that it possible to control, but it is often ignored, not reported, or not accounted for in resistance training practice and research.

Movement tempo is often described as the pace or cadence of a resistance exercise (92). The movement tempo in resistance exercises is usually described using a sequence of digits (e.g., 2/0/X/0), where each digit defines the duration of a particular phase of the movement. Because there is no standardized method of assigning these digits within the scientific literature, in this article, we attempt to unify the description of tempo using the following terms and four-digit combination: eccentric, isometric, concentric, and isometric. For example, 2/0/X/0 denotes a 2-second eccentric phase, no intentional isometric pause during the transition phase, a maximum possible movement tempo during the concentric phase, and no pause between the completion of the concentric phase and the beginning of the next eccentric phase. Additionally, because there is no uniform terminology defining the value of movement tempos, we have used the following criteria in this article (Table 1).

Despite the countless number of scientific studies that focus on resistance training, only a select few have analyzed the effect of different movement tempos during isotonic exercise. Changes in movement tempo at a given external load can influence acute exercise volume, and in turn, the resultant changes in maximum strength, power, and hypertrophy (46,47,52,56,74,88,117,119). Therefore, the main focus of this review is to analyze the current state of knowledge about the influence of movement tempo on acute neuromuscular, hormonal, and mechanical responses to resistance training.

Literature Search

Google Scholar, MEDLINE, ADONIS, ERIC, SPORTDiscus, EBSCOhost, and PubMed databases were searched for all studies investigating the tempo of movement. The search was performed using the following keyword combinations: ("tempo of movement" OR "velocity of movement" OR "repetition duration" OR "speed movement") AND ("time under tension" OR "volume" OR "eccentric duration" OR "concentric duration" OR "endocrine responses" OR "blood" OR "electromyographic"). The present review includes studies that (a) presented original research

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Table 1
Movement tempo classification.

Designation	Abbreviation	Duration of one entire repetition (eccentric, isometric, concentric, isometric)
Volitional	VOL	Volitional speed (V/0/V/0)
Explosive	EXP	Maximum speed possible with that load at that time (X/0/X/0)
Fast	FAS	2–4.9 s
Medium	MED	5–9.9 s
Slow	SLO	10–14.9 s
Extremely slow	ESL	15 s and above

data on healthy adult subjects, (b) were published in peer-reviewed journals, (c) were published in the English language, and (d) used isotonic resistance exercise during the experimental procedures. No sex restrictions were imposed during the search stage.

Impact of Movement Tempo on Acute Muscle Performance

Impact of Movement Tempo on Maximal Load. Resistance training loads are often prescribed using 1 of 2 main methods: a load that corresponds to a certain %1RM or a load where a lifter can perform a maximal number of repetitions within a certain range such as a 10–12RM (7,13). Although these % 1RM and RM range guidelines are commonly used for specific resistance training goals, these guidelines do not exist for or consider specific movement tempos (7,10).

Movement tempo not only depends on the athlete's intent but also, especially in the concentric phase, on the external load used (%1RM), with greater loads decreasing maximal velocity in the concentric phase (74). Along these lines, the use of heavy loads over 85% 1RM will necessitate a near-maximal or maximal effort to concentrically move the load, yet the actual concentric velocity of the lift will be relatively slow. However, in the eccentric phase, even when the external load used is more than 100% 1RM, conscious control of the movement tempo is possible to a certain extent (101). Furthermore, concentric maximal velocity will be reduced even further as a set approaches the point of concentric failure because of the inability of working muscle fibers to maintain force output, which then affects the overall movement tempo. Therefore, even prescribed movement tempos may vary depending on the load used and the amount of fatigue present.

In contrast to what many practitioners may believe, that a slower eccentric phase during a 1RM assessment allows for increased control of the barbell and possibly more active motor units that could increase concentric performance, the opposite may in fact be true. One study examined the effect of 2 tempos (2/0/2/0 and 4/0/2/0) on the maximal 1RM load during the bench press (47). In that study, the maximum load was 3.7% greater during the 2/0/2/0 tempo than during 4/0/2/0, indicating that even during a 1RM assessment, which is assessed via the limitations of the concentric phase, increasing the eccentric duration negatively affects concentric performance. To date, there is no data regarding the differences in 1RM load between fast (FAS) and slow (SLO) or extremely slow (ESL) movement tempos. Nevertheless, considering that extending the eccentric phase by 2 seconds (2/0/2/0 vs. 4/0/2/0) resulted in a 3.7% decrease 1RM, it can be hypothesized that further increasing the duration of the eccentric phase may further decrease the 1RM load compared with a faster movement tempo. Therefore, slower eccentric contractions during a 1RM may not result in any potentiation for the concentric phase but may actually result in a greater physiological demand (65), which could decrease 1RM performance. However, this notion is speculative and should be investigated in future research. Furthermore, it can be assumed that changing the movement tempo affects 1RM performance in both multijoint exercises and isolated movements, but this also requires further research.

In summary, when a controlled movement tempo is used during research or training, 1RM testing should be performed independently for particular tempos and for each exercise. As such, the %1RM value should be determined from the 1RM test trial performed at a specific movement tempo.

Impact of Movement Tempo on the Number of Repetitions Performed. To achieve a desired training volume, the number of repetitions performed per set, per exercise, and per session plays a large role. Although the load used during a resistance exercise largely influences the maximal number of repetitions that can be performed (48,52), the duration of each repetition likely does not stay constant when fatigue ensues. Over the course of an entire training session, this inadvertent change in fatigue and repetition duration can affect the overall training volume. For example, Mookerjee and Ratamess (78) demonstrated that although subjects attempted to perform the concentric phase of repetitions as fast as possible, the duration in the first concentric phase of a 5RM bench press took 1.2 seconds, whereas the fourth and fifth repetitions took 2.5 and 3.3 seconds, respectively. If the same were to occur during a real training session over multiple exercises, the "sets × repetition × load" training volume may not change in the early periods of a training session, but the fatigue induced by additional time under tension (TUT) can largely increase during the latter stages of a training session, which could negatively affect the subsequent number of repetitions.

Some studies have demonstrated that the number of possible repetitions decreases as the total duration of each repetition increases when using the same load (88,117,119). For example, Wilk et al. (117) showed that when using 70% 1RM during 5 sets of the bench press, increasing the duration of movement tempo (2/0/2/0; 5/0/3/0; 6/0/4/0) significantly reduced the maximal number of repetitions performed. They found that the maximal number of repetitions performed in the first set was greater when using a 2/0/2/0 tempo (7 ± 1 reps) compared with 5/0/3/0 (4 ± 1 reps) and 6/0/4/0 (4 ± 1 reps). Additionally, at the end of the 5 sets, the total number of repetitions performed was also greater when using 2/0/2/0 (28.32 ± 6.86 reps) compared with 5/0/3/0 (18.75 ± 4.14 reps) and 6/0/4/0 (15.71 ± 4.03 reps). However, it should be noted that the number of total repetitions between the 5/0/3/0 and 6/0/4/0 tempos was not different, which indicates that the 2-second difference in the duration of movement was not enough to decrease or increase the number of repetitions during 5 sets of bench press at 70% 1RM in a group of experienced resistance-trained men (117). Similarly, Sakamoto and Sinclair (88) showed that a faster movement tempo increased the maximal number of repetitions performed during bench press at 40, 50, 60, 70, and 80% 1RM. In that study (88), 4 different tempos were used: one medium (MED) tempo of 2.8/0/2.8/0, one FAS tempo of 1.4/0/1.4/0, another FAS tempo of 1/0/1/0, and an explosive (EXP) tempo of X/0/X/0 (X represents maximum possible tempo of movement). Similar to the previously discussed study, this study also showed that in general, the maximal number of repetitions performed declined as movement tempo slowed down. However, the unique finding of this study is that the number of repetitions

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was not different between the 2 FAS protocol and the EXP protocol, indicating that similar to the study by Wilk et al. (117), small changes in tempo likely do not affect the maximum number of repetitions.

Rather than using the same load for all protocols, Hatfield et al. (46) investigated the effect of movement tempo on the number of repetitions performed at 60 and 80% 1RM for the shoulder press and back squat exercises. They found that a single set performed with a volitional (VOL) tempo resulted in more repetitions compared with an ESL (10/0/10/0) tempo at both loads and exercises. In contrast, a study conducted by Arazi et al. (6) did not show statistically significant differences between the maximum number of repetitions performed during knee extension exercises (4 sets) with tempos of 3/0/3/0 (50% 1RM), 1/0/1/0 (80% 1RM), and 4/0/2/0 (80% 1RM). In this case, the difference in load used likely played a role in the same number of repetitions being performed (50% 1RM for 3/0/3/0 and 80% 1RM for 1/0/1/0 and 4/0/2/0). Therefore, tempo and load are not mutually exclusive, and they both play a role when it comes to the maximal number of repetitions performed.

Although these studies have clearly shown that slower movement tempo leads to a decrease in the number of possible repetitions with a given load, the study designs used make it impossible to determine precisely whether the number of repetitions is more affected by the duration of the concentric or eccentric phase, or if both phases equally affect the number or repetitions. Only Wilk et al. (119) and Headley et al. (47) have analyzed the impact of changing the duration of only the eccentric phase on the maximum number of repetitions performed in a single strength exercise. For example, the study of Wilk et al. that was described above (119) showed that increasing the eccentric phase 3-fold (6/0/2/0 vs. 2/0/2/0) reduced the number of repetitions by about one third (18 ± 3 vs. 28 ± 4 reps, respectively) during 5 sets of bench press with 70% 1RM. Similarly, in the study by Headley et al. (47), subjects were able to complete more repetitions with 75% 1RM when they used a 2/0/2/0 (6.4 ± 0.6 reps) tempo compared with 4/0/2/0 tempo (4.7 ± 0.4 reps), indicating that extending the eccentric phase again decreased the total number of repetitions possible. Therefore, studies have been consistent in indicating that increasing eccentric duration decreases the maximal number of repetitions, but no studies have sought to identify the minimum changes in duration of movement tempo that would significantly impact the maximal number of repetitions performed in a set. Additionally, more research isolating the effect of only changing the concentric phase is needed to determine how changing the concentric duration affects the number of repetitions performed across a variety of loads and exercises.

From a mechanistic standpoint, a more efficient and more frequent use of the stretch-shortening cycle may be a factor leading to the greater number of repetitions performed at faster tempos, as the stretch-shortening cycle allows for greater muscle force and acceleration during the concentric phase compared with movements without an eccentric component (21,23,24,79). Furthermore, a faster movement tempo, without pausing between the eccentric and concentric phases, can lead to increased concentric contraction velocity and power output compared with a slower tempo, which may be attributed to more efficient utilization of the stretch-shortening cycle (116).

Considering that intentionally slower movement tempos generally decrease the number of repetitions performed and limit the frequency and efficiency of utilization of the stretch-shortening cycle (24,116), strength and conditioning professionals may choose to avoid intentionally slower tempos that decrease the number of repetitions and the utilization of the stretch-shortening cycle because maximizing both is often desired when training for athletic competition. However, there may be some specific situations where sacrificing the number of repetitions and stretch-shortening cycle involvement is not a concern for certain exercises, whereby slower tempos may be more activity specific for athletic performance that requires long periods of constant TUT such as grappling or wrestling an opponent (Table 2). Nevertheless, increasing the duration of eccentric and concentric likely results in earlier concentric fatigue and fewer repetitions performed (70,92) regardless of the type of exercise used.

Impact of Movement Tempo on Time under Tension. Considering the relationship between movement tempo, the number of repetitions, and the TUT, the 3 are not mutually exclusive, as they all affect each other (110,111). Nevertheless, TUT is an indicator of effort and amounts to the total sum of the concentric, eccentric, and isometric components of a repetition (92), meaning that movement tempo directly affects the TUT per repetition. However, as movement tempo affects the number of repetitions that one can perform, changing the movement tempo can indirectly alter the total TUT during a training session, a training microcycle, or even a training mesocycle. Therefore, it is critical to understand how changing movement tempo not only directly affects individual repetitions but also how these small changes can amount to larger changes over time.

The impact of movement tempo on TUT can be determined using simple calculations (e.g., 5 reps with a tempo 2/0/2/0; 5 reps × 4 seconds = 20 seconds TUT). Logically, a slower tempo will result in greater total TUT when the same number of repetitions is performed. For example, Burd et al. (15) compared the effects of 2 different movement tempos on TUT but with equal number of repetitions performed in each set. The subjects performed leg extensions with SLO (6/0/6/0) and FAS (1/0/1/0) movement tempos. In the SLO movement tempo, leg extensions were performed until concentric failure, whereas in the FAS movement tempo, the same number of repetitions was performed with the same load but not until concentric failure. Therefore, the TUT was longer for each set during SLO (set 1 = 198 ± 10 seconds; set 2 = 119 ± 9 seconds; set 3 = 90 ± 7 seconds) compared with FAS (set 1 = 25 ± 2 seconds; set 2 = 14 ± 1 second; set 3 = 11 ± 1 second), resulting in greater total TUT for SLO (407 ± 23 seconds) compared with FAS (50 ± 3 seconds). However, when sets are performed until concentric failure, significant differences may also occur in the total TUT depending on the movement tempo and total number of repetitions completed.

In the study of Wilk et al. (117) where a 6/0/4/0 tempo resulted in fewer repetitions compared with 5/0/3/0 and 2/0/2/0 movement tempos during 5 sets of the bench press to failure using 70% 1RM, the maximal TUT differed significantly in every set between each tempo. In fact, despite resulting in the fewest number of repetitions, the 6/0/4/0 tempo resulted in the greatest total TUT (178.8 ± 33.69 seconds) compared with the 5/0/3/0 tempo (166.60 ± 29.27 seconds) and 2/0/2/0 tempo (124.65 ± 33.66 seconds). Therefore, despite decreasing the training volume (sets × repetitions × load), the slowest tempo resulted in the greatest total TUT, which is important to consider.

Similar results were observed by Hatfield et al. (46) who showed that the changes in TUT may depend not only on the movement tempo but also on the load and type of exercise used. In this study, the total TUT after exercise with VOL tempo was significantly shorter than that after an ESL tempo (10/0/10/0) but only in the shoulder press at 60% 1RM (33.92 ± 4.88 vs. 55.56 ±

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Table 2

Complex resistance training program with different movement tempos of a professional UFC athlete (top 5 in light heavyweight ranking).*

Order	Exercise	Sets	Repetitions	Load (kg)	% 1RM	Tempo	TUT (s)	Rest (s)
1	Bench press	1	15	20	15%	2/0/2/0	60	0
2	Pull-ups	1	10	0	0	X/0/X/0	~10	60
3	Bench press	1	10	60	45%	2/0/2/0	40	0
4	Pull-ups	1	10	0	0	X/0/X/0	~10	60
5	Bench press	1	8	80	60%	2/0/2/0	32	0
6	Pull-ups	1	10	0	0	X/0/X/0	~10	60
7	Bench press	1	6	100	75%	2/0/2/0	24	0
8	Pull-ups	1	10	0	0	X/0/X/0	~10	60
9	Bench press	1	4	110	82%	2/0/2/0	16	60
10	Bench press	2	20	70	52%	X/0/X/0	80	120
11	Power clean	1	10	20	20%	X/0/X/0	~15	120
12	Power clean	1	6	40	40%	X/0/X/0	~10	120
13	Power clean	3	5	60	60%	X/0/X/0	~15	180
14	Standing military press	1	10	20	20%	4/0/X/0	~50	60
15	Standing military press	1	6	40	40%	4/0/X/0	~30	90
16	Standing military press	2	5	50	50%	4/0/X/0	~25	120
17	Standing military press	1	15	40	40%	X/0/X/0	~20	60
18	Barbell squat	1	10	20	13%	2/0/2/0	40	60
19	Barbell squat	1	8	60	40%	5/0/5/0	80	90
20	Barbell squat	1	6	80	53%	5/0/5/0	60	120
21	Barbell squat	1	5	100	67%	5/0/5/0	50	120
22	Barbell squat	3	4	120	80%	5/0/5/0	40	160
23	Barbell squat	1	15	80	53%	X/0/X/0	~20	120
24	One leg dumbbell deadlift	3	10	5	20%	5/0/5/0	100	30
25	Bent-over barbell row	1	10	20	20%	5/0/2/0	70	60
26	Bent-over barbell row	1	10	50	50%	5/0/2/0	70	60
27	Bent-over barbell row	2	10	70	70%	5/0/2/0	70	60
28	Crunches	5	10	0	0	5/0/5/0	100	30

*1RM = 1-repetition maximum; TUT = time under tension.

32.83 seconds, respectively). Importantly, there were no significant differences in the total TUT between tempo ESL and VOL in the shoulder press at 80% 1RM and in the squat at 60 and 80% 1RM, which indicates that the difference in the maximal TUT between the ESL and VOL tempo likely depends on the type of exercise and load used. However, it should be noted that there is no certainty that the duration of VOL movement tempo was the same at each stage. Although no fault of the authors, the nature of VOL tempos make it difficult to control and compare findings within subjects and between subjects, as changes in total TUT can occur by modifying the duration of the eccentric phase, concentric phase, or both. A study by Wilk et al. (119) showed significant differences in maximal TUT in every set of the bench press and in total training TUT between the MED (6/0/2/0) and FAS (2/0/2/0) movement tempos. The maximal total TUT at tempo 2/0/2/0 was 112.59 ± 17.76 seconds, whereas in the tempo 6/0/2/0, it increased to 148.60 ± 25.85 seconds. However, the maximum TUT during resistance training with different movement tempo may be related to the gender (115) and level of experience in resistance training (71,72).

In summary, the changing the movement tempo can have a significant effect on TUT for each set and the entire training session. Naturally, when the number of repetitions is the same, slower movement tempos result in greater total TUT, but this is also likely the case even when performing sets to failure with slower tempos, resulting in fewer repetitions performed. This leads to the conclusion that not only the number of repetitions but also TUT is an important variable to consider when evaluating training volume and optimizing acute responses to resistance training.

Impact of Movement Tempo on Muscle Activity. Changes in movement tempos, external loads, the number of repetitions performed, and the total TUT collectively affect the amount of effort and acute fatigue experienced while resistance training, likely influencing motor unit recruitment, which can be assessed via electromyography (EMG). Electromyography represents the electrical properties of the muscle and is often used to monitor central drive because of the relationship between the amplitude of the surface EMG and the net motor activity (76,99,110). Considering that concentric and eccentric actions have different characteristics in force production and EMG responses (29,49), investigating the possible effects of manipulating movement tempo can provide additional information that could allow for a differentiated prescription of resistance training programs to match the specific needs of an individual. Eccentric and concentric contractions have different EMG signal magnitudes (EMG amplitude) (29,30,107), with lower EMG amplitudes for eccentric actions when compared with concentric actions with the same external load (31). Therefore, altering the concentric or eccentric tempo independently yet equally may not result in similar motor recruitment patterns.

During a training session, a decrease in EMG amplitude may occur between the first and last sets of exercises (33,41,110), when there is presumably more fatigue, evidenced by a reduced integrated EMG (iEMG) signal (103). Previous research has observed that the magnitude and source of such fatigue may vary when different contraction types, intensities, and number of performed repetitions are used (33,81,110). Sakamoto and Sinclair (89) compared changes in muscle activation under different movement tempos and different loads during the bench press

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(40–80% 1RM) performed with muscle failure. Four different tempos were used. The MED tempo was 2.8/0/2.8/0, one FAS tempo was 1.4/0/1.4/0, another FAS tempo was 1/0/1/0, and the EXP tempo was X/0/X/0. The EMG recorded from the pectoralis, deltoid, and triceps brachii showed that the EMG amplitude was greater for the faster movement tempo and heavier load compared with the slower tempo and lower load. However, the faster movement tempo produced a significant decrease in EMG amplitude during the final concentric phase compared with the slower movement tempo. It was concluded that reduced muscle activation during the final concentric phase during the faster movement tempo was related to muscle failure. Similar results were reported by Sampson et al. (90) who showed that faster eccentric elbow flexion produced greater EMG amplitude when compared with slower, longer eccentric actions. During the faster eccentric actions, it is possible that contractile mechanisms would increase force generation because of a higher level of activation (increase in the fraction of cross-bridges formed) during the preactivation period (11). On the contrary, van den Tillaar (108) did not find any differences in muscle activity (semimembranosus, biceps femoris, gastrocnemius, quadriceps) during the squat performed with different movement tempos in the eccentric phases. However, in that study (108), the movement tempo was not precisely defined. Arazi et al. (6) compared the effect of knee extension exercises (4 sets, maximal number of repetitions, one minute rest intervals) with the tempo 3/0/3/0 (50% 1RM), 1/0/1/0 (80% 1RM), and 4/0/2/0 (80% 1RM) on muscle activity measured before and immediately after the last set. Each exercise protocol significantly decreased the iEMG signal between baseline and postexercise, yet there was no significant difference between the conditions that used different movement tempos, despite the fact that the TUT was higher during the slower tempo 4/0/2/0 compared with the 3/0/3/0 and 1/0/1/0 tempo.

The studies described above have analyzed muscle activation while performing resistance training protocols characterized by different load, movement tempo, and number of performed repetitions (6,89,90). However, when manipulating the movement tempo while maintaining the same number of repetitions, one study found that the slower movement tempo resulted in a greater EMG response compared with the faster one (15), and the greater TUT of the slower tempo likely altered EMG amplitude (54,55). Using a similar approach, another study investigated the EMG responses during exercise with different movement tempo, but with equalized TUT or volume load (110). In that study, subjects performed 3 sets of 10 repetitions with 2/0/5/0 and 2/0/2/0 tempos but performed 3 sets of 5 repetitions with a 4/0/10/0. The authors found that all 3 protocols resulted in significant decreases in postexercise iEMG and were not significantly different from each other, which is consistent with Arazi et al. (6). On the contrary, Lacerda et al. (65) showed that despite the fact that the training protocol consisted of equal TUT (36 seconds), muscle activity in the Smith machine bench press exercise was higher during the FAS tempo of movement compared with the MED tempo. Training with MED (3/0/3/0) tempo of movement consisted of 6 repetitions, whereas training with the FAS tempo (1.5/0/1.5/0) consisted of 12 repetitions. Lacerda et al. (65) showed that training protocols conducted with the same TUT, but with a greater number of repetitions and a faster movement tempo, may be a more appropriate strategy to increase muscle activity. Increased muscle activity during resistance exercise with a FAS tempo of movement can be explained by greater peak force generation needed to accelerate the bar when higher movement speed are necessary, thus requiring greater motor unit recruitment (89). This acceleration demand could occur at the beginning of the concentric muscle action. However, when the EMG responses are analyzed with different movement tempos but with equalized TUT, it should be noted that faster movement tempo impact on the the higher repetition numbers performed compared with slower tempo (88,117) that can provoke greater EMG amplitude.

This leads to the conclusion that during the set of resistance exercise, not tempo of movement and not type of exercise but the number of repetitions and TUT is a main factor influencing on EMG responses.

Other Acute Muscle Responses to Resistance Training With Different Movement Tempos

Impact of Movement Tempo on Ratings of Perceived Exertion. Ratings of perceived exertion (RPE) have become a topic of interest in strength training research. In relation to the tempo of movement, Egan et al. (28) compared RPE evaluated after resistance training with the VOL tempo of movement consisting of 6 sets of 6 repetitions of squats at 80% 1RM to the ESL tempo (10/0/10/0) consisting of 6 sets of 6 repetitions using 55% 1RM. These authors found that the training protocol using a VOL movement tempo and high load produce the same RPE compared with the protocol using ESL movement tempo and lower loads. Similarly, Hatfield et al. (46) did not show differences in the RPE between protocols with the VOL tempo of movement and ESL tempo (10/0/10/0) and using the squat and shoulder press exercises (load of 60 and 80% 1RM). On the contrary, Diniz et al. (25) demonstrated that strength training protocols matched by the number of sets and repetitions, load, and rest interval (3 sets; 6 repetitions; 60% 1RM; 3 minutes rest intervals) but with different tempo of movement (4/0/2/0, 2/0/2/0, V/0/V/0) (V represents volitional tempo of movement) produced different responses in RPE. Resistance training with a tempo of 4/0/2/0 yielded greater RPE compared with tempo 2/0/2/0; V/0/V/0; however, no difference in RPE was observed between the 2/0/2/0 and V/0/V/0 tempo. However, it should be noted that in the volitional tempo of movement, there was a high variability of TUT in particular repetitions (1.8–5.3 seconds). This may have contributed to the absence of differences in RPE when compared with the 2/0/2/0 tempo. The literature has already pointed out that a greater number of repetitions can increase the RPE response (82) and it may explain part of the results obtained by Hatfield et al. (46), but there is no available data regarding the relationship between TUT and RPE.

Impact of Movement Tempo on Cardiovascular Responses. There are few studies investigating the effects of different movement tempos on cardiovascular responses to resistance exercises. Tanimoto and Ishii (104) continuously measured blood pressure during knee extensions with tempos of 3/1/3/0 (50% 1RM), 1/1/1/0 (50% 1RM), and 1/1/1/0 (80% 1RM). They showed that peak systolic blood pressure (SBP) during the 1/1/1/0 tempo at 80% 1RM was significantly higher than that during the 3/1/3/0 and 1/1/1/0 tempos at 50% 1RM, which indicates that the tempo did not have a major impact on changes in blood pressure during resistance exercise, but the external load used did affect blood pressure, with the heavier load increasing blood pressure compared with the lighter load.

Resistance exercise can not only induce blood pressure changes during training but also reduce the postexercise blood pressure to a level below what it was preexercise. This phenomenon is called postexercise hypotension (PEH) (57,69) and has been widely

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investigated because of its importance for the treatment and prevention of arterial hypertension (45,84). However, in regard to different movement tempo, only one study (5) examined PEH responses to different tempos during resistance exercise. The exercise protocol of Arazi et al. (5) consisted of 3 sets of 8 repetitions at 80% 1RM (leg press, lat pull-down, knee flexion, bench press, knee extension, and standing biceps extension) performed with a 1/0/1/0, 1/0/2/0, or 2/0/4/0 movement tempo. After completing each training session, SBP and diastolic blood pressure (DBP) were taken every 10 minutes during 60 minutes of recovery. The results indicated significant increases in SBP at 10th min postexercise in comparison to baseline. After 60-minute recovery, all conditions showed statistically significant decreases in SBP when compared with preexercise values. At all time points, there were no significant differences in SBP and DBP among the considered exercise tempos of movement.

Lastayo et al. (66) demonstrated that the eccentric cadence uses up to 6–7 times less oxygen compared with concentric. Furthermore, studies have shown that the concentric phase contributes the most to the O_2 uptake during the lift and recovery, with the eccentric phase consuming very little O_2 during a resistance exercise (26,93). The results described by Dudley et al. (26) provided the basis for a metabolic explanation of the higher TUT in the ESL and SLO tempos compared with FAS, especially when the changes are associated with the extension of the eccentric phase of the movement (66,93). Considering the small amount of research on this topic, further research should be conducted before definitive statements can be made regarding the effect of different movement tempos on the cardiovascular responses to resistance exercise.

Impact of Movement Tempo on Delayed Onset Muscle Soreness. Only one study has compared the effects of different movement tempo on delayed onset muscle soreness (DOMS) and its associated indications (4). Arazi and Chegini (4) analyzed different eccentric tempos (4/0/1/0, 3/0/1/0, 2/0/1/0, 1/0/1/0) and evaluated serum creatine kinase, lactate dehydrogenase, and muscular pain perception before, 24 hours after, and 48 hours after exercise. There were no changes for lactate dehydrogenase, but creatine kinase and pain perception in all groups at 24 and 48 hours after resistance exercise had a significant increase. A significant difference between groups 1/0/1/0 and 2/0/1/0 to 4/0/1/0 group was observed. Therefore, it seems that slower movement tempo can result in greater muscle damage and DOMS compared with faster tempos, but more research should be performed in this area before conclusive statements can be made.

Impact of Movement Tempo on Metabolic and Hormonal Blood Responses

Recent studies have suggested that acute metabolite accumulation during and after resistance training may be the primary stimulus for gains in strength and muscle hypertrophy (17). The key hormones in training adaptations are testosterone (particularly in men), growth hormone, and cortisol. Another important indicator of blood responses to stress induced by resistance training is blood lactate levels. Those hormones and blood lactate have an effect on the changes occurring after a resistance training program because of interaction with specific androgen receptors (64). Previous research that analyzed different movement tempo and number of repetitions showed that changes in these variables affects physiological responses such as hormones and blood lactate concentrations (85,104,112,119). It has been suggested that increasing the duration of repetition without changing the number of repetitions performed could increase the metabolic response provided by resistance training (77,112).

Metabolic Responses: Blood Lactate. The blood lactate response during and after exercise is used in different modalities of sport to determine training workloads (106). Numerous studies have analyzed changes in blood lactate levels following resistance training, yet few of them concerned changes in blood lactate in response to different movement tempos.

A study by Hunter et al. (52) compared the effects of ESL (5/0/10/0) and FAS (1/0/1/0) movement tempos on postexercise concentrations of blood lactate. Although the duration of training sessions in this research was equal for both tempos used (29 minutes), blood lactate was significantly greater for FAS compared with ESL (7.9 ± 1.7 vs. 4.0 ± 2.0 $mmol \cdot L^{-1}$ blood). Greater blood lactate levels after training with FAS movement tempo occurred despite nearly 3 times lower total TUT after training with FAS compared with SLO movement tempo (256 and 960 seconds, respectively). Furthermore, a study by Pareja-Blanco et al. (83) showed that blood lactate was significantly higher after training with EXP compared with half EXP tempo at 60, 70, and 80% 1RM, although TUT was significantly shorter for training with EXP tempo. Arazi et al. (6) showed an increase in postexercise blood lactate concentration after 4 sets of knee extensions, but the group that preformed the exercise with the 3/0/3/0 tempo at 50% 1RM had significantly lower lactate concentration than the group that performed the same exercise at the 1/0/1/0 and 4/0/2/0 tempo with 80% 1RM. However, it should be noted that the difference in the load used between these tempos (50 and 80% 1RM) could have had a significant impact on the obtained results.

These results reflected greater metabolic stress following training with faster compared with slower movement tempos. The authors argued that greater force generation would be required to perform faster movements, which would result in greater recruitment of muscle fibers with higher glycolytic potential despite longer TUT in training with slower movement tempo.

In contrast, Wilk et al. (119), Martins-Costa et al. (70), and Mazzetti et al. (73) showed that the training with slower movement tempo and longer TUT resulted in greater increases in blood lactate measurements compared with faster tempo and shorter TUT. Specifically, the study by Mazzetti et al. (73) compared 2/0/2/0 tempo with 128 seconds TUT and 2/0/X/0 tempo with 96 seconds TUT, whereas the study by Wilk et al. (119) compared SLO (6/0/2/0) and FAS (2/0/2/0) movement tempos. After 5 sets of bench press exercise with maximal number of repetitions, blood lactate levels were greater in the group using SLO movement tempo compared with FAS (119). In the study by Martins-Costa et al. (70), despite the fact of equal value of training parameters, except for difference in movement tempo (2/0/2/0 vs. 4/0/2/0), higher blood lactate levels were present after MED (4/0/2/0) compared with FAS (2/0/2/0). However, TUT during MED tempo was longer compared with FAS (108 vs. 72 seconds). The differences in TUT values can be explained by higher concentration of blood lactate after a longer effort with slower movement tempo. These data support the argument that slower repetitions increase some components of workout intensity (e.g., total contraction volume relative to rest intervals) (54,114,119).

Other results, published by Headley et al. (47), Keogh et al. (58), and Tanimoto et al. (105), did not show differences in the postworkout concentration of blood lactate between resistance

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training with different movement tempos. Keogh et al. (58) compared training with VOL, EXP, and SLO movement tempo. Despite significant differences in load and values of TUT in tempos, there were no significant differences in postworkout value of blood lactate between VOL, EXP, and SLO movement tempos. In a study by Tanimoto et al. (105), changes in blood lactate levels during exercise were similar after 3/0/1/0 and 1/0/1/0 tempos despite much lower intensity used in 3/0/1/0 (55–60% 1RM) compared with 1/0/1/0 (80–90% 1RM). Similarly, Headley et al. (47) did not show any differences in blood lactate levels after using a 2/0/2/0 compared with a 4/0/2/0 tempo. However, it should be noted that those studies were not matched by training volume, TUT, or training load and that differences in these components may also interfere with blood lactate responses (14,65).

Only a few scientific studies have analyzed the impact of training with different movement tempos on blood lactate but with equal TUT. Lacerda et al. (65) compared blood lactate responses after training with FAS tempo (1.5/0/1.5/0) and MED tempo (3/0/3/0) with equal TUT (108 seconds). The study showed that blood lactate concentrations were higher after training with FAS tempo (1.5/0/1.5/0), compared with MED tempo (3/0/3/0), which indicates that the number of repetitions performed is a more stimulating factor for increasing blood lactate. Interesting research was made by Goto et al. (38) who compared blood lactate concentrations with an equal number of repetitions, TUT, and load but with a different movement tempo. The research showed higher blood lactate concentrations after 1/0/5/0 compared with 5/0/1/0. Longer duration during the concentric phases denied arguments that greater force generation would be required to perform faster movements, which result in a greater recruitment of muscle fibers with higher glycolytic potential. The greater lactate production in response to concentric contractions persists even when the same relative workload is applied (59).

The different blood lactate responses in presented studies following resistance training with different movement tempo may be related to the mechanical characteristics of the protocols, type of exercise used (multijoint or one-joint), the size of muscle area activation during the effort, and the value of external load used. Furthermore, considering that more repetitions are performed using faster movement tempos compared with slower tempo, higher maximal forces would be expected to accelerate the bar during every repetition performed with faster tempo (46). With the generation of higher maximal forces in protocols with faster tempo, additional motor units with higher glycolytic capacities were presumably recruited (14,47,83), which might promote an increase in blood lactate production after faster compared with slower tempo of movement. The results are in agreement with the findings of Tran and Docherty (110), which showed that when equivalent in TUT, protocols performed with higher number of preformed repetitions led to increased levels of fatigue (reduced ability to generate force), indicating a greater physiological demand during its execution. In contrast, Wilk et al. (119) showed that training with slower movement tempo increases maximal value of TUT, whereas longer work time leads to higher blood lactate levels. However, the research published by Hunter et al. (52) and Pareja-Blanco et al. (83) demonstrated that longer duration of TUT is not the main factor influencing the changes in blood lactate concentration, even when the slower movement tempo feels more intense and causes greater fatigue compared with a faster tempo. Although slower tempo during the resistance exercise used is very demanding and fatiguing, it did not entail greater contraction intensity or exercise intensity, and thus, it did not result in the fastest rate of energy expenditure (52).

Hormonal Responses. Several studies have also investigated the relationship between acute hormone responses following different resistance training protocols. There is a consensus that the endocrine responses to resistance exercise depending on training characteristics, such as the amount of muscle mass activated, the exercise order, the load used, the number of repetitions performed per set, the number of sets per exercise, and the length of rest interval between sets (3,77,95,96). Movement tempo is another variable that may affect the biological stress of resistance training and thus the endocrine response.

Testosterone. Resistance exercise protocols of high volume (3–6 sets; 8–12 reps), moderate load (60–85% 1RM), and short rest intervals (30–90 seconds), which activate large muscle groups, elicit the greatest acute elevations in testosterone (2,8,12,17,39,68,75,96). However, there are no guidelines regarding the tempo of movement in these recommendations.

Smilios et al. (97) compared different movement tempos, consisting of 4 sets of squats and 4 sets of leg press exercises using a load of 10RM. EXP movement tempo or 70% of the EXP tempo of movement were used in the experiment. Testosterone concentration increased following all protocols, which is in agreement with previous studies (75,86). However, no differences were observed in testosterone concentration between the 2 exercise protocols at any time point. Headley et al. (47) also did not find any differences in testosterone responses between MED (4/0/2/0) and FAS (2/0/2/0) tempo of movement, although a significantly greater amount of repetitions were performed in the FAS tempo of movement compared with the SLO one (6.4 vs. 4.7). However, the TUT was, contrary to the number of repetitions performed, significantly higher for the SLO movement tempo compared with the FAS one (28.2 vs. 25.6 seconds).

In contrast, Wilk et al. (119) and Goto et al. (40) found greater testosterone responses after training with the slower movement tempos. Goto et al. (40) compared MED (3/0/3/0) and FAS (1/0/1/0) tempos but with different loads. The MED tempo at 40% 1RM caused significantly greater free testosterone responses than the FAS tempo at 80% 1RM and 40% 1RM. The differences, especially those in the loads applied, may explain the discrepancies in the results of Goto et al. (40). A study by Wilk et al. (119) compared SLO (6/0/2/0) and FAS (2/0/2/0) movement tempos during 5 sets of the bench press exercise at equal load (70% 1RM). Every set in the SLO and FAS tempo was performed to concentric muscular failure. The study showed that postexercise levels of testosterone were greater after the SLO protocol compared with the FAS one. However, after 30 and 60 minutes of recovery, there were no differences in testosterone level between exercise protocols. Goto et al. (38) compared knee extensions performed with 1/0/5/0, 5/0/1/0, and 3/0/3/0 at 50% 1RM and 1/0/1/0 at 80% 1RM. The results indicated that low load (50% 1RM) with slower movement tempo increased testosterone concentrations most significantly. The findings published by Goto et al. (38) and Wilk et al. (119) suggest that extending the duration of particular repetitions and TUT is an important factor in stimulating testosterone responses to resistance exercise, regardless of the load used. However, when training sessions were equalized for tempo of movement and TUT, significantly greater elevations in postexercise testosterone were registered after 3 shorter sets compared with one longer one (37). Furthermore Cintineo et al. (20) analyzed testosterone responses after resistance exercise with 1/0/1/0 tempo consisting of 3 sets compared with 4/0/2/0 tempo and only 1 set. Both groups had equal TUT of approximately 60 seconds. Testosterone increased from pre- to postexercise and

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remained elevated at 30 minutes after exercise in both the tempos (1/0/1/0 and 4/0/2/0), although there were no significant differences between the groups. Another study compared straight sets with no rest between repetitions with a protocol with a 30-second intraset rest interval and found no deviations from baseline at any time point with the exception of 30-minute postexercise, where the no-rest group showed a significant decrease in testosterone, although this was not significantly different from the other group (37) and may be indicative of increased receptor binding.

Therefore, both volume (number of repetitions, as well as the TUT), load used, number and distribution of sets (118), and the type of exercise are important factors in determining the magnitude of testosterone response to resistance exercise (119), which limits the possibility of indicating one main factor affecting significant postexercise testosterone changes.

Growth Hormone. Optimal training variables stimulating significant postexercise elevations in growth hormone (hGH) concentration are similar to those for testosterone. In a study by Smilios et al. (97), hGH increased immediately after resistance exercise and remained elevated for 20 minutes into recovery in all exercise protocols compared with the baseline values. Furthermore, immediately after exercise, hGH was higher following the 70% EXP tempo of movement versus the EXP one. The statistically greater hGH response in this study was observed after training with 70% EXP tempo of movement where the subjects made a higher training volume (by 10.6% than in the EXP tempo of movement). Therefore, a higher training volume favors a greater hGH response, which confirms previous findings (16,36,96). Goto et al. (40) also found statistical greater hGH responses after training with MED compared with FAS tempo of movement (3/0/3/0 vs. 1/0/1/0). Importantly, hGH responses were greater for exercise with low load (40% 1RM) and MED tempo compared with high load (80% 1RM) and FAS tempo. Interestingly, the total volume of training basic on number of repetitions in the group using the FAS tempo was approximately twice as high as the in the SLO group, which undermines previous indications that the volume of training is a factor determining the postexercise increase in hGH concentration. However, total volume basic on TUT was significantly higher during SLO compared with FAS. In similar studies, the same authors compared knee extensions performed with 1/0/5/0, 5/0/1/0, and 3/0/3/0 at 50% 1RM and 1/0/1/0 at 80% 1RM. hGH concentration increased to a greater extent after the 3 slower movement tempos compared with the faster movement tempo (38). In contrast, Headley et al. (47) did not find any statistical differences in hGH responses between groups using FAS (2/0/2/0) and MED (4/0/2/0) tempos of movement despite the fact that significantly higher values of TUT were observed in the group using slower tempo. Goto et al. (37) compared hormonal responses between subjects performing a single set and 3 sets, yet with equal TUT and equal tempo of movement. Significantly greater elevations in postexercise hGH were found after the exercise protocol consisting of 3 sets compared with the single set, despite the fact that TUT and the tempo of movement were equalized.

Changes in hGH responses depend on training volume. Higher volume protocols with greater metabolic requirements lead to increased hGH responses after resistance training (36,96). However, it should be noted that when the optimal volume is exceeded, the hGH peak can occur already during the training session (113,118), which was not recorded or analyzed in most previous studies. It has been speculated that local accumulation of anaerobic energy metabolites, such as lactate, stimulate the hypophysis secretion of hGH (60,102). Consequently, a slower movement tempo may result in greater metabolic stress, resulting in increased hGH secretion after a bout of resistance exercise.

Table 3

Training sessions with different movement tempos.*

Order	Exercise	Tempo	Sets/repetitions	TUT (s)
1	Barbell squat	5/0/5/0	3/5	150
2	Barbell deadlift	2/0/2/0	3/12	144
3	Leg extension	2/0/1/0	3/15	135
4	Lying leg curls	5/0/5/0	3/5	150

*TUT = time under tension.

Cortisol. Cortisol, as a stress hormone, increases after the execution of higher volume protocols combined with high metabolic stress, as compared with lower volume protocols (22,75,96). The results published by Smilios et al. (97) showed that cortisol levels after resistance training with maximal tempo of movement were higher at all postexercise time points compared with the 70% EXP tempo of movement. Furthermore, Goto et al. (38,40) found a higher cortisol increase during resistance exercise performed with slower compared with faster tempo. In a study by Goto et al. (40), researchers compared MED (3/0/3/0) and FAS (1/0/1/0) tempos using a low load (40% 1RM). The study by Goto et al. (38) compared 1/0/5/0, 5/0/1/0, and 3/0/3/0 tempos at 50% 1RM and 1/0/1/0 at 80% 1RM. Cortisol levels were significantly higher after 5/0/1/0 tempo compared with 1/0/5/0 and 1/0/1/0 tempos, whereas the 1/0/5/0 tempo exhibited the greatest increase in cortisol 15 minutes after exercise. Therefore, the low load exercise with slower eccentric phases stimulated smaller changes in cortisol compared with low load exercise with slower concentric phases. Cintineo et al. (20) compared cortisol responses between resistance exercise with a 1/0/1/0 (3 sets) and 4/0/2/0 (1 set) tempo of movement but with equal TUT of approximately 60 seconds. Cortisol increased from pre to post exercise and remained elevated 30 minutes into recovery with the 1/0/1/0 tempo (3 sets), whereas after the 4/0/2/0 tempo (1 set), the tested subjects exhibited no change in cortisol from baseline at any time point. Because cortisol is a catabolic hormone that mobilizes substrates during exercise, these results further indicate that the amount of induced metabolic stress was significantly greater in the 1/0/1/0 exercise protocol compared with the 4/0/2/0 protocol.

In contrast, the studies by Headley et al. (47) and Wilk et al. (119) did not find any effect of movement tempo on cortisol responses when only the duration of the eccentric phase was modified. Headley et al. (47) compared 2/0/2/0 and 4/0/2/0 tempos at 75% 1RM, whereas Wilk et al. (119) compared 2/0/2/

Table 4

Resistance exercises performed with different movement tempos.*

Set	Exercise	Tempo	Load (% 1RM)	Repetitions (n)	TUT (s)
1st	Barbell squat	V/0/V/0	20%	15	~60
2nd	Barbell squat	2/0/2/0	50%	10	40
3rd	Barbell squat	2/0/2/0	80%	8	32
4th	Barbell squat	6/0/2/0	80%	4	32
5th	Barbell squat	2/0/2/0	80%	8	32
6th	Barbell squat	6/0/2/0	80%	4	32
7th	Barbell squat	5/0/X/0	50%	15	~90

*1RM = 1-repetition maximum; TUT = time under tension.

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Table 5
Training microcycle with a different movement tempos.

Training sessions	Monday	Tuesday	Wednesday	Thursday	Friday
Tempo/muscle area	Slower tempo for upper-muscle area	Faster tempo for lower-muscle area	Day -off	Faster tempo for upper-muscle area	Slower tempo for lower-muscle area

0 and 6/0/2/0 using a load of 70% 1RM. In both studies, the exercise was performed to concentric failure. However, the maximal number of repetitions performed and TUT was significantly different between both tempos. In the study by Headley et al. (47), subjects were able to complete more repetitions when they used the 2/0/2/0 compared with the 4/0/2/0 tempo. Similarly, a higher number of repetitions were completed at a faster tempo in a study by Wilk et al. (119). In contrast to the number of repetitions performed, a longer TUT was found for training with slower movement tempo (47,119).

It should be noted that cortisol responses were greatest after 4–6 sets compared with just 2 sets (96) and after 3 sets compared with 1 set (20), indicating that not the tempo of movement, but high amounts of metabolic stress, may be the main factor influencing postexercise cortisol concentrations. The acute cortisol response has been shown to be correlated with long-term adaptations to resistance training (113); thus, one may speculate that the protocol consisting of several sets of a resistance exercise may induce greater muscle remodeling than single longer set exercise protocols. Furthermore, the research showed that when the resting concentration of C is high, no postexercise elevation was found (8), and in some cases, a postexercise decrease in concentrations of this hormone was observed compared with resting levels (118). The results of research also indicate that there are significant differences in individual cortisol responses to specific types of exercise (8).

Conclusion

The results presented in this review indicate that movement tempo affects the total TUT and number of performed repetitions, which subsequently affect endocrine responses following resistance exercise. However, differences in the size of the muscles examined, the structure of the training programs, and the experimental approach used may partially explain the discrepancy in results between the faster and slower tempo of movement. Furthermore, the total TUT and the ratio of concentric to eccentric duration during each repetition may also be important factors to consider because concentric actions produce greater increases in blood lactate concentrations, sympathetic nerve activation (18), RPE (38), motor unit recruitment (29), and cardiovascular responses (18) following constant external resistance exercise. Considering that during the eccentric contraction, the energy requirements are typically 4-fold smaller than that during the concentric phase of movement with the same load (50), it would seem logical that lower metabolic and hormonal stress would occur. This should affect the metabolic and endocrine responses when compared with concentric-only contractions even when TUT is equalized. However, studies indicate that the duration of the eccentric phase of movement can be widely manipulated during resistance training (92,94). From a practical point of view, using a fast but controlled duration of the eccentric phase (~2 seconds) may allow for time-efficient training sessions. However, extending the duration of the eccentric contraction, yet with a simultaneous decrease in the number of repetitions performed, will not cause excessive lengthening of the training session. Another method that can be used during resistance training with controlled movement tempo based on eccentric contractions includes accentuated eccentric loading (62) or the use of eccentric only contractions with a supramaximal load (above 100% 1RM) (35,61). The use of weight releasers allows for overloading the muscles during the eccentric phase of movement because of its specific construction. The weight can be unloaded in the transition from the eccentric to the concentric phase of movement. The use of high loads during the eccentric phase of movement and slower movement tempo is associated with significant exercise-induced muscle damage and mechanical tension, which have been associated with significant hypertrophic and strength responses (62,91).

Although limited research exists on sex differences in resistance training, recent reports suggest that using different strategies for men and women may be more effective when striving at improvement of strength, power, and hypertrophy (9,32). However, currently there is no empirical data on intersexual differences in acute effects of resistance training with different movement tempo. On average, women typically possess 60–80% of the strength, muscle fiber, and whole muscle anatomical cross-sectional area of men (27,80,109). The lower blood androgen levels of women also have been hypothesized to respond with less relative strength and muscle hypertrophy to resistance exercise when compared with men (44). However, for lower-body training, a number of studies have failed to find any difference between male and female subjects with similar relative improvements, both in terms of hypertrophic and strength adaptive changes after resistance exercise (1,51,87). On the contrary, differences in jumping ability (19,98,100), jumping kinematics/kinetics (98,100), musculoskeletal stiffness characteristics (42,43,63), and the effective use of eccentric jump component (19,98) have been observed between male and female subjects. However, none of these studies considered the tempo of movement used during the resistance exercise. The review of Folland and Williams (32), Laubach (67), and Garhammer (34) led to a conclusion that women have a lower capacity during slower, strength-oriented lower-body activities than in faster, power-oriented lower-body

Table 6
Impact of movement tempo on exercise volume.*

Tempo	2/0/1/0	5/0/3/0	8/0/4/0	10/0/5/0
1RM (from 2/0/1/0)	140	140	140	140
% 1RM	80%	60%	50%	40%
Load (kg)	112.5	85	70	55
Number of sets (n)	5	5	5	5
Number of repetitions (n)	6	6	6	6
TUT per repetition (s)	3	8	12	15
Set × Rep (n)	30	30	30	30
Set × Rep × TUT (s)	90	240	360	450
Load × Set × Reps (kg)	3,375	2,550	2,100	1,650
Load × Set × Reps × TUT (kg × s)	10,125	20,400	25,200	24,750

*1RM = 1-repetition maximum; TUT = time under tension.

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Table 7

Summary of studies exploring the influence of movement tempo on acute changes to resistance training.*

Reference	Tempo/Load (%1RM)	Subjects	Protocol	Main findings
Arazi et al. (5)	1/0/1/0 (80% 1RM) 1/0/2/0 (80% 1RM) 2/0/4/0 (80% 1RM)	13 resistance-trained females	Leg press, lat pull-down, knee flexion, bench press, knee extension, biceps extension 3 sets × 8 reps	↔ SBP and DBP between tempos
Arazi et al. (6)	1/0/1/0 (80% 1RM) 3/0/3/0 (50% 1RM) 4/0/2/0 (80% 1RM)	24 physically active men not participated in a regular resistance training program	Knee extension 4 sets × maximal number of reps to exhaustion	↔ Maximal REP between tempos ↔ iEMG between tempos ↑ Post-exercise level of blood lactate in 3/0/3/0 compared with 1/0/1/0 and 4/0/2/0
Burd et al. (15)	1/0/1/0 (30% 1RM) 6/0/6/0 (30% 1RM)	8 resistance-trained men	Unilateral knee extension 3 sets × 6 reps	↓ TUT during tempo 1/0/1/0 compared with 6/0/6/0 ↑ EMG for slower tempo
Calixto et al. (16)	3/−/−/− 0,5/−/−/− (70% 1RM-ECC)	12 resistance-trained men	Bench press 4 sets × 8 reps (ECC only)	↑ Post-exercise level of hGH in 0.5/−/−/− compared with 3/−/−−
Cintineo et al. (20)	1/0/1/0 (100% 10RM) 4/0/2/0 (75–80% 10RM)	19 resistance-trained men	Leg press, leg extension, Romanian deadlift, prone leg curl, seated adductor, seated calf raise 1/0/1/0–3 sets in exercise 4/0/2/0–1 set in exercise Equal TUT	↔ Post-exercise level of testosterone between tempos and time point ↑ Post-exercise level of testosterone for 1/0/1/0 compared with 4/0/2/0
Diniz et al. (25)	V/0/V/0 (60% 1RM) 2/0/2/0 (60% 1RM) 4/0/2/0 (60% 1RM)	12 resistance-trained men	Bench press on a Smith machine 3 sets × 6 reps	↑ RPE after 4/0/2/0 compared with V/0/V/0 and 2/0/2/0 ↔ RPE between V/0/V/0 and 2/0/2/0
Egan et al. (28)	V/0/V/0 (80% 1RM) 10/0/10/0 (55% 1RM)	14 resistance-trained women	Squat 6 sets × 6 reps	↑ RPE after V/0/V/0 compared with 10/0/10/0
Goto et al. (38)	1/0/5/0 (50% 1RM) 5/0/1/0 (50% 1RM) 3/0/3/0 (50% 1RM) 1/0/1/0 (80% 1RM)	12 resistance-trained men	Knee extension 4 sets × maximal number of reps to exhaustion	↑ Post-exercise level of blood lactate in 1/0/1/0 compared with others tempos ↑ Post-exercise level of blood lactate in 5/0/1/0 compared with 1/0/5/0 ↑ Post-exercise level of cortisol in 5/0/1/0 compared with 1/0/5/1 and 1/0/1/0
Goto et al. (40)	1/0/1/0 (40% 1RM) 1/0/1/0 (80% 1RM) 3/0/3/0 (40% 1RM)	12 resistance-trained men	4 sets × maximal number of reps to exhaustion	↑ Post-exercise level of hGH and testosterone in 3/0/3/0 compared with 1/0/1/0 at 40 and 80% 1RM ↔ Post-exercise level of cortisol between tempos
Hatfield et al. (46)	V/0/V/0 10/0/10/0 (60%; 80% 1RM)	9 resistance-trained men	Squat and shoulder press 1sets × maximal number of reps to exhaustion	↑ Maximal REP during tempo V/0/V/0 compared with 10/0/10/0 in squat and shoulder press at every load ↓ TUT during tempo V/0/V/0 compared with 10/0/10/0 in shoulder press at 60%1RM ↔ TUT between tempos in shoulder press at 80% 1RM and in squat at 60 and 80% 1RM ↔ RPE between tempos in shoulder press at 80% 1RM and in squat at 60 and 80% 1RM
Headley et al. (47)	2/0/2/0 4/0/2/0	17 resistance-trained men	Part 1: 1RM test for both tempos Part 2: Bench press; 4 sets; First set: 4 reps at 55% 1RM Second set: 5 reps at 60% 1RM Third set: 6 reps at 65% 1RM Fourth set: maximal number of reps to exhaustion at 75% 1RM	Part 1: ↓ 1RM test during tempo 4/0/2/0 compared with 2/0/2/0 Part 2: ↔ Post-exercise level of lactate, testosterone, human growth hormone, cortisol between tempos
Hunter et al. (52)	1/0/1/0 (65% 1RM) 5/0/10/0 (25% 1RM)	8 resistance-trained men	10 exercise 1/0/1/0: 2 sets × 8 reps 5/0/10/0: 1 set × 8 reps Equal duration of training session—29 min	↑ Post-exercise level of blood lactate in 1/0/1/0 compared with 5/0/10/0

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Table 7

Summary of studies exploring the influence of movement tempo on acute changes to resistance training.* *(Continued)*

Reference	Tempo/Load (%1RM)	Subjects	Protocol	Main findings
Keogh et al. (58) 1999	V/0/V/0 (80% 1RM) X/0/X/0 (30% 1RM) 5/0/5/0 (55% 1RM)	12 resistance-trained men	Squat V/0/V/0–1 set × 6 reps X/0/X/0–1 set × 6 reps 5/0/5/0–1 set × to maintain constant velocity	↔ Post-exercise blood lactate level between tempos
Lacerada et al. (65)	1.5/0/1.5/0 (60% 1RM) 3/0/3/0 (60% 1RM)	12 resistance-trained men	Smith machine bench press 1.5/0/1.5/0–3 sets × 12 reps 3/0/3/0–3 sets × 6 reps Equal TUT	↑ Post-exercise level of blood lactate in 1.5/0/1.5/0 compared with 3/0/3/0 ↑ EMG during the 1.5/0/1.5/0 compared with 3/0/3/0
Martins-Costa et al. (70)	2/0/2/0 (60% 1RM) 4/0/2/0 (60% 1RM)	15 recreationally trained men	Bench press 3 sets × 6 reps	↑ Post-exercise level of blood lactate in 4/0/2/0 compared with 2/0/2/0 ↑ TUT in tempo 4/0/2/0 compared with 2/0/2/0
Mazzetti et al. (73)	2/0/X/0 (60% 1RM) 2/0/2/0 (60% 1RM)	9 resistance-trained men	Squat 4 sets × 8 reps	↑ Post-exercise level of blood lactate in 2/0/2/0 compared with 2/0/X/0
Pareja-Blanco et al. (83)	X/0/X/0 50% X/0/X0 (60; 70; 80% 1RM)	21 resistance-trained men	Squat 3 sets × 8 reps at 60% 1RM 3 sets × 6 reps at 70% 1RM 3 sets × 3 reps at 80% 1RM	↑ Post-exercise level of blood lactate in X/0/X/0 compared with 50% X/0/X/0 at every load
Sakamoto and sinclar (88)	2.8/0/2.8/0 1.4/0/1.4/0 1/0/1/0 X/0/X/0 (40; 50; 60; 70; 80% 1RM)	13 resistance-trained men	Bench press on a Smith machine 1 set × maximal number of reps to exhaustion at every load	↓ Maximal REP when duration of tempo increased
Sakamoto and sinclar (89)	2.8/0/2.8/0 1.4/0/1.4/0 1/0/1/0 X/0/X/0 (40; 50; 60; 70; 80% 1RM)	13 resistance-trained men	Bench press on a Smith machine. 1 set × maximal number of reps to exhaustion at every load	↑ EMG amplitude for faster tempos and heavier load
Sampson et al. (90),	X/0/X/0 (100% 6RM) 2/0/X/0 (100% 6RM) 2/0/2/0 (100% 6RM)	12 recreationally trained men	Elbow flexion and extension 1 set × maximal number of reps to exhaustion	↔ Maximal REP between tempos ↑ TUT during the 2/0/2/0 compared with 2/0/X/0 and X/0/X/0 ↑ EMG amplitude for faster tempo
Smilios et al. (97)	X/0/X/0 (10-RM) 70% X/0/X/0 (10-RM)	6 resistance-trained men	Squat, leg press 4 sets × 8 reps	↔ Post-exercise level of testosterone between tempos ↑ Post-exercise level of hGH in 70% X/0/X/0 compared with X/0/X/0 ↑ Post-exercise level of cortisol in X/0/X/0 compared with 70% X/0/X/0
Tanimoto and Ishii (104)	1/1/1/0 (50% 1RM) 1/1/1/0 (80% 1RM) 3/1/3/0 (50% 1RM)	24 untrained men	Knee extension 3 sets × maximal number of reps to exhaustion	↑ SBP during the 1/1/1/0 (80% 1RM) compared with 1/1/1/0 (50% 1RM) and 3/1/3/0 (50% 1RM)
Tanimoto et al. (105)	1/0/1/0 (80–90% 1RM) 3/0/1/0 (55–60% 1RM)	9 resistance-trained men	Squat, bench press, latissimus dorsi pull-down, abdominal bend, and back extension 1 set × maximal number of reps to exhaustion	↔ Post-exercise blood lactate level between tempos after each type of exercise
Tillaar et al. (108)	Self-selected slow, normal, and fast eccentric movement (4-RM)	11 resistance-trained men	Squat 1 set × 4 reps	↔ EMG activity between tempos
Tran and Docherty (110) 2006	2/0/5/0 (90% 10RM) 2/0/2/0 (90% 10RM) 4/0/10/0 (90% 10RM)	10 resistance-trained men	Elbow flexion 3 sets × 10 reps for 2/0/5/0 and 2/0/2/0 3 sets × 5 reps for 4/0/10/0	↔ iEMG between tempos
Wilk et al. (115)	2/0/X/0 (70% 1RM) 6/0/X/0 (70% 1RM)	20 resistance-trained women	Close-grip bench press Wide-grip bench press 5 sets × maximal number of reps to exhaustion	↓ Maximal TUT during tempo 2/0/X/0 compared with 6/0/X/0 ↓ TTUT during tempo 2/0/X/0 compared with 6/0/X/0 ↑ Maximal REP during tempo 2/0/X/0 compared with 6/0/X/0 ↑ TREP during tempo 2/0/X/0 compared with 6/0/X/0 ↔ REP, TUT between grip width

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Table 7

Summary of studies exploring the influence of movement tempo on acute changes to resistance training.* *(Continued)*

Reference	Tempo/Load (%1RM)	Subjects	Protocol	Main findings
Wilk et al. (117)	2/0/2/0 (70% 1RM) 5/0/3/0 (70% 1RM) 6/0/4/0 (70% 1RM)	42 resistance-trained men	Bench press 5 × maximal number of reps to exhaustion	↓ maximal TUT during tempo 2/0/2/0 compared with 5/0/3/0 and 6/0/4/0 ↓ TTUT during tempo 2/0/2/0 compared with 5/0/3/0 and 6/0/4/0 ↑ maximal REP during tempo 2/0/2/0 compared with 5/0/3/0 and 6/0/4/0 ↑ TREP during tempo 2/0/2/0 compared with 5/0/3/0 and 6/0/4/0
Wilk et al. (119)	2/0/2/0 (70% 1RM) 6/0/2/0 (70% 1RM)	16 resistance-trained men	Bench press 5 sets *x* maximal number of reps to exhaustion	↓ Maximal TUT during tempo 2/0/2/0 compared with 6/0/2/0 ↓ TTUT during tempo 2/0/2/0 compared with 6/0/2/0 ↑ Maximal REP during tempo 2/0/2/0 compared with 6/0/2/0 ↑ TREP during tempo 2/0/2/0 compared with 6/0/2/0 ↑ Post-exercise level of blood lactate and testosterone in 6/0/2/0 compared with 2/0/2/0 ↔ Post-exercise cortisol level between tempos

*Tempo of movement = eccentric/isometric/concentric/isometric; 1RM = 1 repetition maximum; TUT = time under tension; TTUT = total time under tension; REP = repetition; TREP = total number repetition; ECC = eccentric; RPE = ratings of perceived exertion; SBP = systolic blood pressure; DBP = diastolic blood pressure; ↑ denotes significant increases; ↔ denotes no significant differences; ↓ denotes significant decreases; iEMG = integrated electromyography.

activities, which indicates a significant impact of movement tempo during resistance exercise arising from the gender of the subjects. Therefore, future research should consider possible differences between men and woman in acute and chronic responses to resistance exercise with different movement tempo.

Practical Applications

Considering the data discussed throughout this review, movement tempo should be taken into consideration when planning and executing resistance training programs. Changing movement tempo can influence the 1RM, total TUT, and the number repetitions performed. First, during resistance exercise research or training, when a controlled movement tempo is used, 1RM testing should be performed independently for particular tempos. The % 1RM value should be determined from the 1RM test trial performed at a specific movement tempo. Slower movement tempo increases maximal exercise duration, which can be particularly important in speed endurance events like in track and field, swimming, cycling, and combat sports. Furthermore, programming resistance training based on TUT allows to precisely adjust the time of effort in each set to the real time of effort during sports competition. However, the use of a slower tempo of movement may limit the frequency and efficiency of the stretch-shortening cycle, which can be counterproductive in sports requiring explosive movements, such as the jumps and throws in athletics. However, there is a lack of data confirming the negative impact of slower movement tempo on power output, especially when resistance training is just one of the many different forms of training of the competitive athlete. In this case, the coach and athlete can consider using a slower movement tempo only during the eccentric phase of the contraction (which will cause an increase in muscle activation and lengthening the TUT) with an explosive movement in the concentric phase contraction (optimal for development of power output). Another possibility is the use complex training, by combining a resistance exercise performed with a slower tempo of movement, followed by an explosive or plyometric exercise with a biomechanically similar movement pattern. Such a slower, high-tension, plyometric complex training can be an effective alternative compared with traditional resistance training, which could help athletes break through plateaus and prevent training monotony. However, there is no published data assessing the effectiveness of complex resistance training, consisting of resistance exercises performed with a slower movement tempo, followed by plyometric exercises. Nevertheless, unpublished research results from the Strength and Power Laboratory of the Academy of Physical Education in Katowice, Poland, shows that such complex training may be an effective form of improving power output in different sport disciplines (e.g., ski jumping, combat sports, athletic jumps and throws, basketball, volleyball, etc.). We have observed positive effects (unpublished data) of a 6-week training program in a recent study that compared the MED (6/0/X/0) tempo of movement with a VOL one (V/0/V/0) in regards to strength with EXP (V/0/X/0) in power output gains in competitive athletes. Importantly, it must be stated that the slower or controlled movement tempo does not have to be used in every training session, in every set or even in every repetition. It is a common training practice to use alternative tempo of movement in one training

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session (Table 3) or even in sets of an exercise (Table 4). The tempo of movement can also be varied in successive training sessions in microcycle, in an attempt to prevent monotony and overtraining (Table 5). The principles of periodization also apply to the tempo of movement, but currently, it was not proposed by any resistance training authorities, despite the fact that athletes including Olympic champions and world champions have been using such training solutions several years (Table 2). Additionally, a variable tempo of movement can be favorable for youth athletes or for injured athletes who, for various reasons, would be unable to use heavy loads or perform explosive movements. The use of a slower tempo of movement during resistance exercise allows the use of a lower external load compared with a faster tempo and simultaneously increase the TUT during particular sets as well during training session (Table 6). Furthermore, during the slower movement tempo, it is easier to control particular phases of movement, which may be particularly beneficial in the recovery process injured athletes. Therefore, resistance training programs should include information about the used movement tempo. Moreover, slower movement tempos allow for an extension of exercise duration, even if the number of repetitions remains unchanged, which leads to the conclusion that not only the number of repetitions but also the TUT is an important variable in the evaluation of training volume and optimization of acute changes in resistance training (Table 7).

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